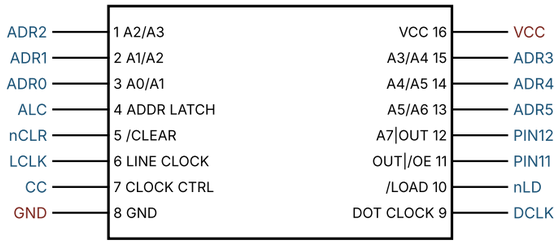
Hazeltine 1420 character-generator adapter, rev B (replaces DM8678 / DM86S64 at U25 / U26)

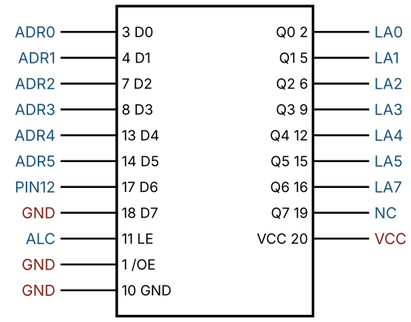
net-label schematic: pins carrying the same label are connected. Two board variants (U26 upper / U25 lower) share this circuit.

P1 plug into DM8678 / DM86S64 socket (U25 / U26)

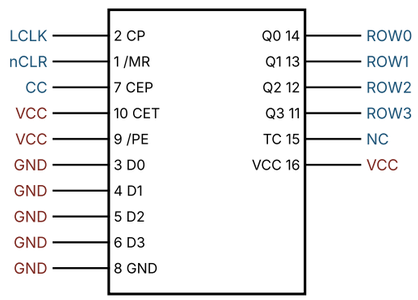


pin 11 = video OUT, pin 12 = A7 / enable, active low
(same on both chips, measured)

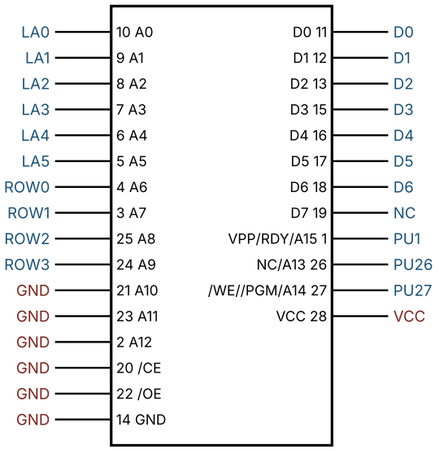
U1 74HCT373 address latch



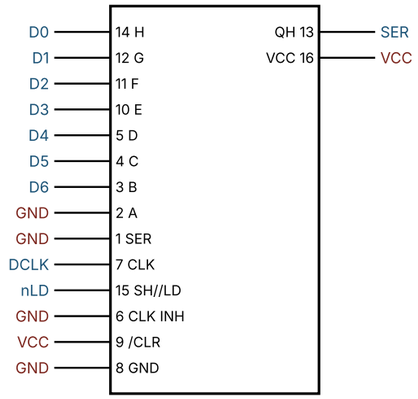
U2 74HCT161 line counter



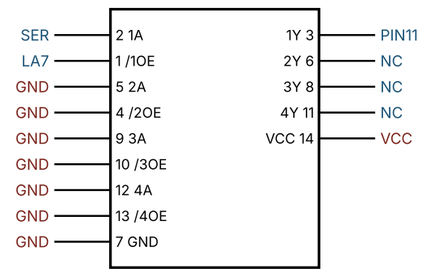
U5 AT28C64B font EEPROM (DIP-28 socket)



U3 74HCT166 dot shift register



U4 74HCT125 tri-state output




```
R1, R2, R3  1 kOhm 0805   VCC -> U5 pin 1, 26, 27  (nets PU1, PU26, PU27): keep /WE high on a 28C64,
            set A13/A14/A15 high on a 27C256/27C512 (image block replicated, see build_eeprom.py)
C1..C5      100 nF 0805 decoupling, one per IC (VCC-GND)

Rev B: no jumpers.  Video out U4 pin 3 -> socket pin 11 (net PIN11); latched socket pin 12 (A7) -> U4 pin 1 /OE (net LA7).
        (Rev A had solder jumpers JP1/JP2 for a DM86S64 pin assignment that turned out to be the same as the DM8678 one;
         rev A boards are used with JP1 C-A and JP2 C-A bridged, which is exactly this wiring.)

Output enable polarity: U4 = 74HCT125 -> output active when latched pin 12 is LOW, as measured on both original chips.
If the font dump of a chip shows it outputs when pin 12 is HIGH, fit a 74HCT126 instead (same pinout, active-high enable).

Logic family: 74HCT (TTL-compatible inputs, VIH = 2.0 V).  The terminal drives the socket with 74LS outputs (VOH >= 2.7 V),
which plain 74HC (VIH = 3.5 V) would not reliably read.  Do not substitute 74HC for U1-U4.

Timing: address -> latch (74HCT373, ~25 ns) -> EEPROM (150 ns) -> 74HC166 setup: ~190 ns < 280 ns spec of the
original (t_S2).  Dot clock up to 25 MHz with 74HCT166; Hazeltine 1420 dot clock is below 20 MHz.

Row address: the 74HC161 replaces the on-chip mod-16 line counter (CP = line clock, CEP = clock control,
/MR = /clear).  Descender shifting of the original is baked into the EEPROM image row by row.
Shift register: /LOAD -> SH//LD; DOT CLOCK -> CLK; EEPROM D0 (first dot) -> H ... D6 -> B; A and SER = 0,
so the output goes low after the 7 dots exactly like the original.
```

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